JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 89, NO. B6, PAGES 4461–4477, JUNE 10, 1984

Northward Displacement and Accretion of Wrangellia: New Paleomagnetic Evidence from Alaska

JOHN W. HILLHOUSE AND C. SHERMAN GROMME

U.S. Geological Survey

Wrangellia is a vast terrane that accreted along the Pacific margin of North America from Oregon to central Alaska. Previous paleomagnetic studies of the Triassic volcanic rocks of Wrangellia in south-central Alaska and British Columbia have given paleolatitudes of 10°–17°, which are anomalously low in comparison to results from similar-aged rocks of nuclear North America. The discrepancy in paleolatitude implies substantial northward drift of the terrane relative to the craton. We have extended paleomagnetic sampling to the northern boundary of the terrane in the Mount Hayes and Healy quadrangles of the central Alaska Range. This collection of 46 Triassic lava flows yields a mean paleolatitude of 13.9° ($\alpha_{95} = 3.8°$); the polarity of the data and hence the hemisphere of origin are not resolved with certainty. In contrast, Triassic paleomagnetic poles from the stable part of North America predict a paleolatitude of 42.1° N ($\alpha_{95} = 4.6°$) for the region. The northernmost fragment of Wrangellia probably reached mainland Alaska in the middle Cretaceous, when Upper Jurassic and Lower Cretaceous flysch within the intervening basin underwent severe deformation. Tertiary transcurrent faulting has modified the positions of Wrangellian fragments in Alaska, but such displacements are probably minor, as indicated by paleomagnetic data from early Tertiary volcanic rocks that overlap the terrane.

INTRODUCTION

Paleomagnetic studies demonstrate that vast tracts of southern Alaska have moved thousands of kilometers north relative to cratonic North America since Triassic time. One of the displaced terranes, called Wrangellia by *Jones et al.* [1977], is defined by a distinctive stratigraphic sequence of upper Paleozoic and lower Mesozoic rocks. Near McCarthy, Alaska, one unit of this sequence, the Triassic Nikolai Greenstone, was sampled for paleomagnetic study [*Hillhouse*, 1977]. Inclinations determined from 50 Nikolai basalt flows yielded a far lower paleolatitude than is predicted for the region on the basis of Triassic continental reconstructions. Reconciliation of the paleomagnetic results with Wrangellia's present position requires that the terrane has moved at least 3000 km poleward relative to the craton. A concurrent paleomagnetic study of basalt which is correlative with the Nikolai in other parts of southern Alaska [*Stone*, 1982] reached a similar conclusion. Here we report new results from additional sampling of the Nikolai Greenstone and correlative basalts in the central Alaska Range near the northwest margin of Wrangellia (Figures 1 and 2).

Beyond south-central Alaska the distinctive rock assemblage of Wrangellia can be traced into isolated, fault-bounded fragments in southeastern Alaska, British Columbia, eastern Oregon, and western Idaho. The recognition of Wrangellia as a widespread and possibly displaced terrane [*Jones et al.*, 1977] grew out of stratigraphic studies by *Berg et al.* [1972], *Monger et al.* [1972], and *Csejtey* [1976]. Before the terrane was recognized, *Symons* [1971] found the first clues to the anomalous paleolatitude of Wrangellia while studying the Triassic Karmutsen Formation of Vancouver Island, British Columbia. He recognized two components of magnetization in the basalt flows and proposed northward displacement of the island as one of several possible explanations for the results. In a concurrent study, *Irving and Yole* [1972] separated the Karmutsen magnetizations into a primary cooling component and an overprint and postulated the northward displacement of Vancouver Island. Additional work on Vancouver Island [*Yole and Irving*, 1980; *Schwarz et al.*, 1980], in Hells Canyon, Oregon [*Hillhouse et al.*, 1982], and in south-central Alaska [*Hillhouse*, 1977] reaffirmed the low Triassic paleolatitudes that were suggested by the earlier studies. Along the entire length of Wrangellia from Oregon to Alaska (approximately 20° of latitude) the paleolatitudes are unexpectedly low and remarkably similar. The implication is that the terrane has undergone a complex history of northward drift, fragmentation, and accretion. Geologic mapping combined with paleomagnetic studies provide clues to this complex history. For example, Triassic sequences in terranes that now border Wrangellia, such as the Alexander and Peninsular terranes, are sufficiently different to imply greater separation of the terranes in the Triassic [*Jones et al.*, 1977]. Paleomagnetic evidence from the Alexander terrane [*Hillhouse and Gromme*, 1980] supports this view. During northward drift, the fragments of Wrangellia were joined by other terranes, as indicated by low paleolatitudes of overlapping sedimentary rocks [*Stone et al.*, 1982; *Monger et al.*, 1982; *Csejtey et al.*, 1978, 1982]. The amalgamation of terranes is also indicated by the existence of crosscutting plutonic belts, such as the Coast Plutonic Complex of British Columbia and early Tertiary granitic rocks of south-central Alaska. In this paper we examine the paleomagnetic and geologic evidence that documents the exotic origin of Wrangellia, marks the amalgamation with neighboring terranes, and constrains the timing of accretion to the continental margin.

This paper is not subject to U.S. copyright. Published in 1984 by the American Geophysical Union.

Paper number 4B0165.

STRATIGRAPHY AND PALEOMAGNETIC SAMPLING OF WRANGELLIA IN THE MOUNT HAYES AND HEALY QUADRANGLES, ALASKA

In south-central Alaska (Figure 2), the distinctive assemblage of upper Paleozoic and lower Mesozoic rocks that defines Wrangellia is limited to the area between the Border Ranges and Denali faults [*Jones et al.*, 1981]. The terrane is recognized as far west as the Talkeetna Mountains, where it terminates against the Talkeetna thrust, a designation which

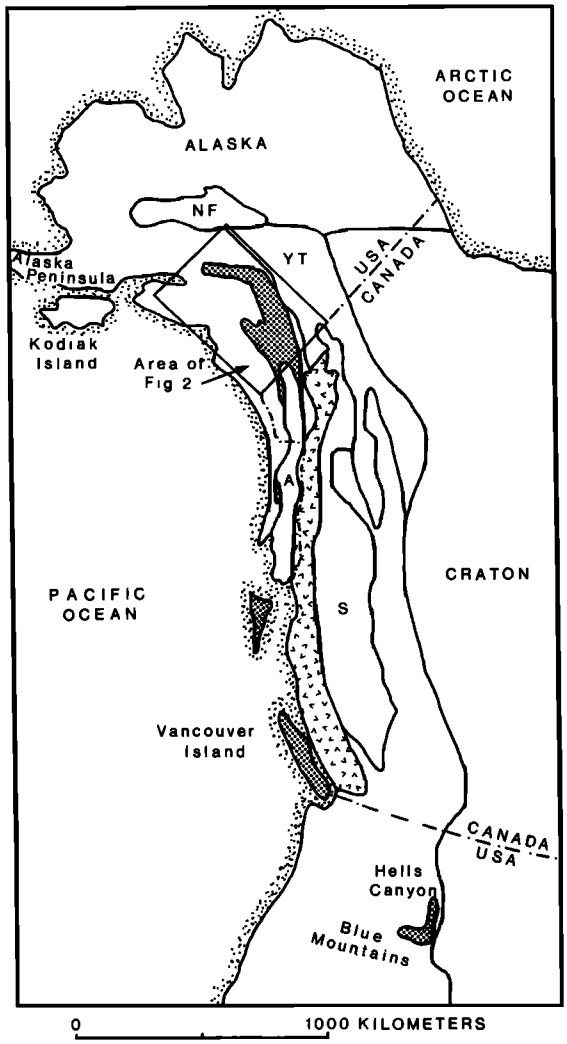


Fig. 1. Index map showing selected tectonostratigraphic terranes of northwestern America, after *Coney et al.* [1980]. Wrangellia is indicated by the stippled pattern; *V* pattern denotes Coast Plutonic Complex of British Columbia. NF indicates Nixon Fork terrane, YT the Yukon Tanana terrane, S the Stikine terrane, and A the Alexander terrane.

includes the Broxson Gulch thrust [*Stout*, 1976; *Csejtey et al.*, 1978, 1982]. Near McCarthy the lower part of the Wrangellian sequence consists of Pennsylvanian and Lower Permian andesitic volcanic and calcareous rocks of the Skolai Group [*Smith and MacKevett*, 1970]. In the Mount Hayes quadrangle, 250 km to the northwest, rocks correlative with the Skolai Group are the Pennsylvanian Tetelna Volcanics and the overlying Mankomen Group [*Richter and Dutro*, 1975]. Similar rocks, described as part of the Amphitheater Group of *Smith and Turner* [1973] by *Smith* [1974] and *Stout* [1976], occur in the Amphitheater Mountains [*Silberling et al.*, 1981*a*].

An unconformity separates Lower Permian volcanogenic rocks from the distinctive Triassic assemblage that characterizes Wrangellia. This assemblage is dominated by the Nikolai Greenstone [*Rohn*, 1900], which at McCarthy consists of at least 3000 m of subaerial basalt flows, thick volcanic conglomerate beds, and minor volcaniclastic rocks [*MacKevett*, 1978]. The Nikolai is underlain by thin, discontinuous lenses of black shale containing Middle Triassic (Ladinian) fossils of the genus *Daonella* at McCarthy [*MacKevett*, 1978] and near the Slana River [*Richter et al.*, 1977]. Volcanic rocks of the Mount Hayes quadrangle that are considered by *Silberling et al.* [1981*a*] to be correlative with the Nikolai Greenstone are part of the Amphitheater Group, as defined by *Stout* [1976], which includes the Boulder Creek Volcanics, the upper part of the Tangle Lakes Formation, and the Paxson Mountain Basalt. Triassic volcanic rocks of the Amphitheater Group [*Smith*, 1981] in the western Clearwater Mountains are also considered correlative with the Nikolai. Throughout south-central Alaska the Triassic basalt flows and underlying strata are intruded by sills, dikes, and large bodies of gabbro that were feeders to the vast volcanic field.

The Nikolai Greenstone is overlain by the Upper Triassic (Karnian) Chitistone Limestone at McCarthy and by correlative unnamed limestones to the northwest. Throughout Wrangellia volcanic rocks did not erupt after the Karnian. *Silberling et al.* [1981*a*] concluded that Norian fossils attributed to limestone lenses within the Amphitheater Group [*Smith and Turner*, 1973; *Smith*, 1981] actually came from younger slivers in fault contact with the volcanic rocks. The upper part of Wrangellia consists of Upper Triassic and Lower Jurassic calcareous marine rocks, as typified by the McCarthy Formation [*MacKevett*, 1978].

In the Mount Hayes quadrangle, Wrangellia rocks have been divided into the Tangle and Slana River subterranes by *Nokleberg et al.* [1981, 1982]. The Slana River subterrane is similar to Wrangellia rocks at McCarthy; important characteristics are that the Nikolai Greenstone consists of subaerial basalts and that the underlying Paleozoic island arc assemblage is relatively thick. The Tangle subterrane is south of the Slana River subterrane, is separated from it by a series of thrust faults, and is distinguished by a relatively thinner Paleozoic section and by a much thicker section of Nikolai Greenstone, the lower part of which is pillow basalt.

From the Nikolai Greenstone we collected samples for

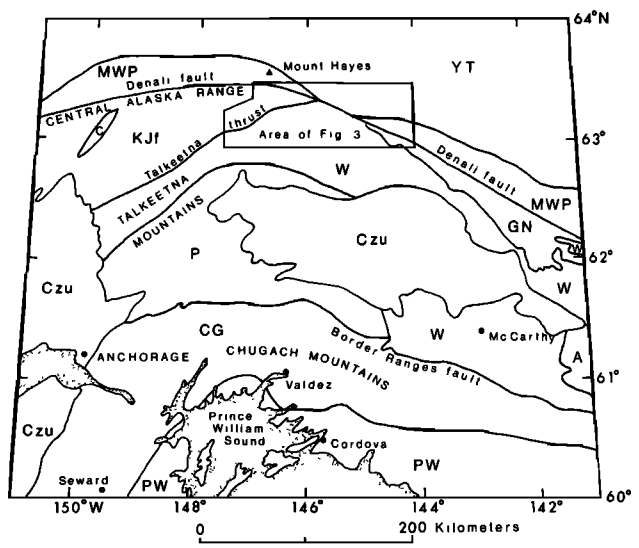


Fig. 2. Tectonostratigraphic terranes and overlap assemblages of southcentral Alaska, after *Jones et al.* [1981]. Terrane designations: A: Alexander, C: Chulitna, CG: Chugach, P: Peninsular, PW: Prince William, W: Wrangellia, YT: Yukon-Tanana, and MWP: McKinley, Windy, and Pingston, undivided. Overlap assemblages: Czu: Cenozoic rocks, undivided, GN: Gravina Nutzotin Belt, KJf: Cretaceous and Jurassic deformed flysch.

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

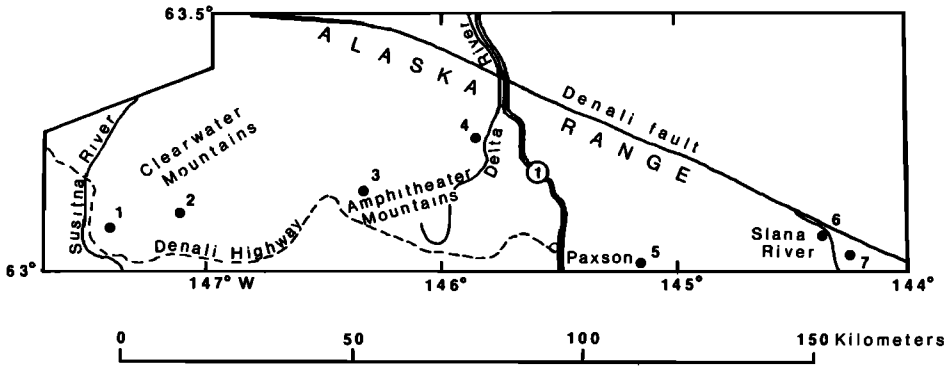


Fig. 3. Location of sites for paleomagnetic study in Triassic basalt of Healy and Mount Hayes quadrangles, Alaska. Site numbers are keyed to Tables 1 and 2.

paleomagnetic study at seven sites, distributed from the Clearwater Mountains to the Slana River, a distance of 150 kilometers in the Mount Hayes and eastern Healy quadrangles (Figure 3). Sites 1, 2, 3, and 5 are in the Tangle subterrane and sites 4, 6, and 7 are in the Slana River subterrane [*Nokleberg et al.*, 1982; *Silberling et al.*, 1981*b*]. The sites were carefully selected for freshness of exposure and for the availability of well-defined bedding to ensure good structural control. Each site is a stratigraphic section composed of six to 11 lava flows (Table 1), forming a total of 46 flows. Generally, we collected eight independently oriented cores from each flow. Azimuths were measured with a magnetic compass, and corrections for local anomalies were made by backsighting to several landmarks.

The flows, 1 to 10 m thick, typically consist of dark green or maroon-gray amygdaloidal basalt, although a volcanic breccia was sampled at site 5. As described by *Richter et al.* [1977], the basalt of sites 6 and 7 contains phenocrysts of saussuritized plagioclase, clinopyroxene, and relict olivine in an intergranular groundmass that has been largely altered to chlorite, epidote, and serpentine. Amygdules are typically zoned and consist of calcite, quartz, potassium feldspar, chlorite, epidote, and zeolite minerals. Minor amounts of native copper occur in the amygdules. Low-grade metamorphism is indicated by the presence of prehnite and pumpellyite. Chemical analyses of the Triassic lavas of the Clearwater Mountains (sites 1 and 2) suggest original compositions in the field of tholeiitic basalt and andesite [*Smith*, 1981]. All samples that were examined in thin section showed preservation of original igneous textures and no evidence of penetrative shearing or deformation.

Demagnetization Methods and Analysis of Magnetic Components

Natural remanent magnetizations (NRM) of the basalt ranged in intensity from 10^{-1} to 10 A/m (1 A/m = 10^{-3} emu/cm^3), and magnetic directions generally showed large within-flow angular dispersions. The large dispersion suggested the presence of multiple components of magnetization at each site. The multicomponent content of NRM was borne out by stepwise alternating field (AF) and thermal demagnetization experiments which were applied to two cores from each lava flow. Progressive demagnetization of the samples was carried out in alternating fields up to 100 mT in a shielded 400-Hz coil. The field was applied successively to three orthogonal axes of the specimen, which was held stationary during each application. Whenever possible, AF and thermal demagnetization techniques were applied to companion specimens cut from the same core so that the effectiveness of the treatments could be compared directly. During thermal demagnetization the specimens were progressively heated in air in a shielded furnace. After cooling in an ambient magnetic field of less than 10 nT, remanent magnetizations of the specimens were measured.

Orthogonal vector diagrams were plotted to depict the decay of remanence during demagnetization, and difference vectors were plotted to show the component that was removed by successive demagnetization steps. The treatments revealed generally two components of magnetization, which were most apparent during thermal demagnetization (Figures 4, 5, and 6). The first is a steep downward component (not

TABLE 1. Paleomagnetic Sample Sites in the Nikolai Greenstone, Central Alaska Range

Site (Figure 3)	Latitude, deg north	Longitude, deg east	Location	Flows Sampled	Section Thickness, m	Structural Attitude	
						Strike, deg	Dip, deg
1	63.09	212.62	Clearwater Mountains	10	80	220	41N
2	63.12	212.89	Clearwater Mountains	11	300	293.3	59.7 N
3	63.16	213.68	Amphitheater Mountains	8	100	238.8	33.8 N
4	63.27	214.13	Rainy Mountain	8	150	112	32S
				2	50	281	31.5 N
5	63.02	214.86	Caribou Lake	4	55		
6	63.09	215.63	Slana River	6	35	298	61N
7	63.05	215.72	Slana River	5	105	293	40N
				4	75	298	32N

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

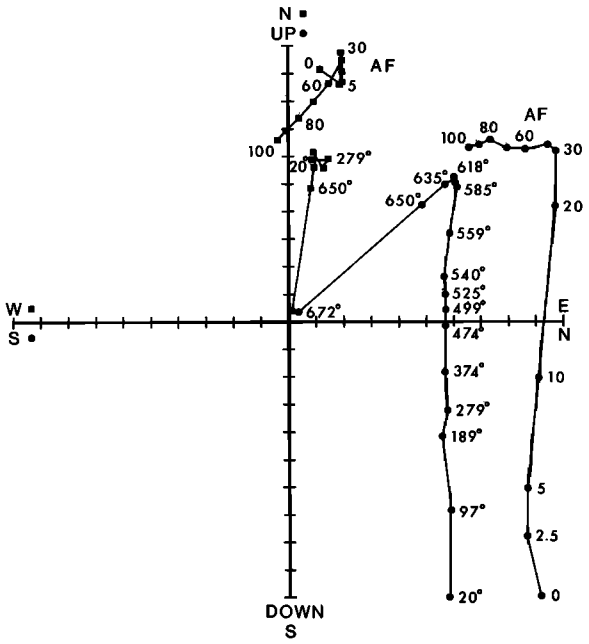


Fig. 4*a*. Orthogonal projections of remanence vector (not corrected for tilt of bedding) during thermal and alternating field (AF) demagnetization of companion specimens from site 1, flow 9 (Figure 3; Tables 1–3). AF values in mTesla; temperatures in °C. Squares are projected into the horizontal plane; circles lie in vertical-north plane. One division = 1.45 $\times 10^{-1}$ A/m for thermal treatment or 1.05 $\times 10^{-1}$ A/m for AF.

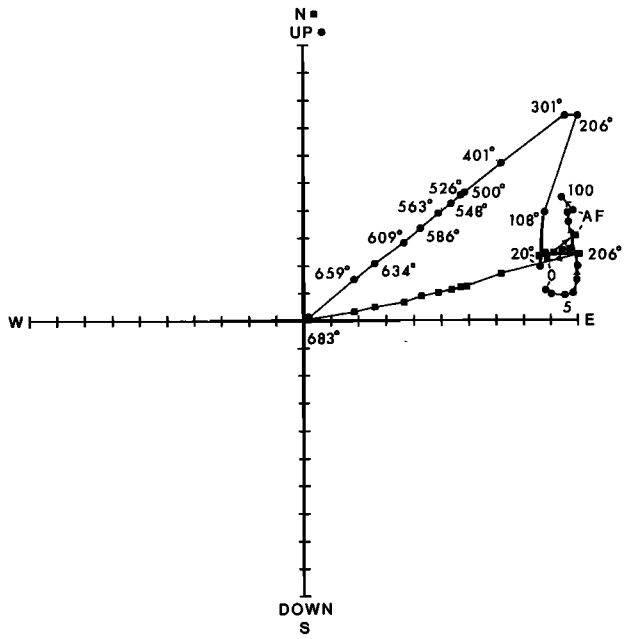


Fig. 5*a*. Orthogonal projections (not corrected for tilt of bedding) from progressive demagnetization of specimens from site 7, flow 6 (Figure 3; Tables 1–3). Circles lie in vertical-east plane. Other symbols as in Figure 4*a*. One division = 2.45 $\times 10^{-2}$ A/m for thermal treatment or 1.07 $\times 10^{-2}$ for AF.

corrected for tilt of the bedding) that gave highly variable unblocking temperature distributions between room temperature and 600°C. The second component generally had higher unblocking temperatures, in the range 300°–680°C, and was directed to the east with shallow upward inclinations (not corrected for tilt of the bedding). Although most specimens exhibited a superposition of both components, which were separable at high temperature, some showed no trace of the shallow component and vice versa.

A comparison of vector diagrams revealed that AF demagnetization was not as effective as thermal treatment in the isolation of the stable endpoint magnetization of most specimens (Figure 5). "Stable endpoint magnetization" is attained when the magnetic intensity decays toward zero while the magnetic direction changes negligibly. AF demagnetization was particularly ineffective in decreasing the intensity of specimens dominated by the shallow upward component of magnetization. Therefore most of the remaining untreated specimens were given only thermal treatment to isolate the stable endpoint magnetization. Exceptions to this procedure were made for sites 2 and 3, where AF treatments of 40 mT were employed. In these cases, AF demagnetization was as effective, if not better, than heating in isolating the endpoint remanence. Some flows from site 2 were apparently affected by lightning, as indicated by anomalously intense magnetizations. These specimens were given a double treatment in which they were first treated in an AF of 40 mT and then heated in the range 550°–575°C.

The thermal treatment of the specimens from each flow was prescribed from visual inspection of the stepwise vector diagrams, which determined the temperature range between iso-

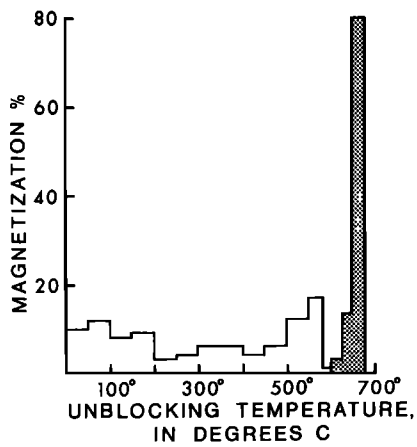


Fig. 4*b*. Unblocking temperature distribution derived from thermal demagnetization of specimen in Figure 4*a*. Secondary component is unblocked between 20° and 618°C; characteristic magnetization (stippled pattern) above 618°C.

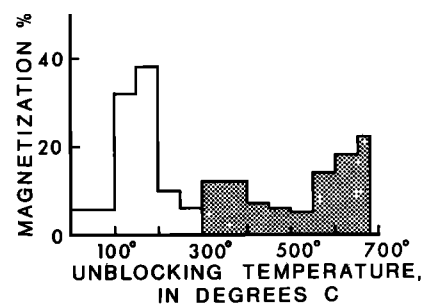


Fig. 5*b*. Unblocking temperature distribution derived from Figure 5*a*. Characteristic magnetization (stippled pattern) is isolated above 301°C.

21562023b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

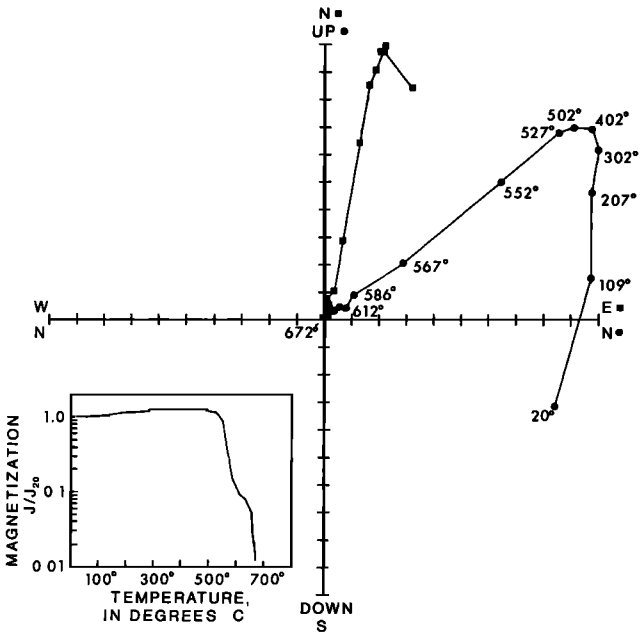


Fig. 6. Orthogonal projections (not corrected for tilt of bedding) after thermal demagnetization of specimen from site 1, flow 1 (Figure 3, Tables 1–3). Symbols as in Figure 4*a*. One division = 7.98 × 10^{-1} A/m. Normalized intensity plot (inset) shows characteristic magnetization is carried by magnetite and hematite.

lation of the stable endpoint remanence and complete demagnetization. All specimens except those noted above were heated to five temperature steps which overlapped this range, usually at 25°C intervals. After plotting these results on vector diagrams, we selected the highest heating step which typified the endpoint magnetization. A few specimens (4%) exhibited erratic changes of direction throughout this treatment, so they were rejected as being magnetically unstable.

For most specimens the steep and shallow components are easily separated by heating the specimens to within 25°C of the total unblocking temperature. In these cases, a plot of the difference vectors obtained by heating the specimens from

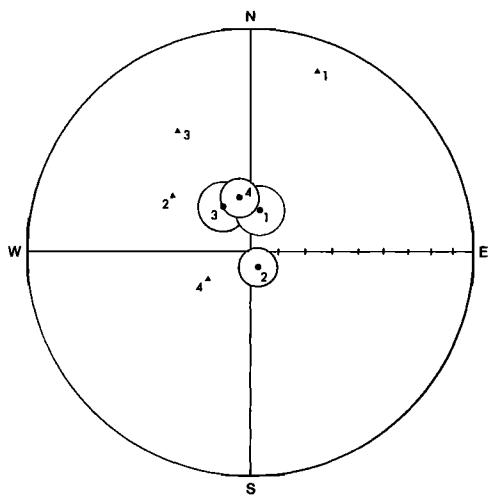


Fig. 7. Magnetic overprint directions and 95% confidence limits (circles) obtained by vector subtraction (200°–300°C) before (dots) and after (triangles) application of tilt corrections. Lower hemisphere, equal-area projection. Site numbers are keyed to Table 1 and Figure 3.

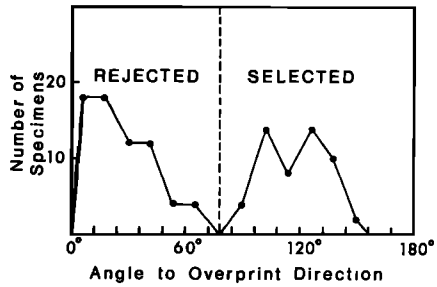


Fig. 8. Histogram showing basis for data selection (site 3).

200° to 300°C shows that the steep component fails the fold test (Figure 7), as angular dispersion increases significantly after correcting directions for tilt of the bedding. We are therefore justified in rejecting this family of directions as a natural overprint that was acquired after deformation of the rocks. In contrast, application of the tilt correction to the shallow component does not significantly increase or decrease the directional dispersion. The shallow component roughly parallels the strikes of the bedding, so a successful fold test is precluded. Nevertheless, the shallow component is the dominant endpoint magnetization that resides at generally higher blocking temperatures than does the steep component. As explained in a following section, we interpret the shallow component to be the original magnetization acquired during cooling of the basalt.

We employed the following procedure to separate the completely overprinted endpoint magnetizations from the shallow endpoint magnetizations. For each site, difference vectors that were obtained after heating pilot specimens from 200°–300° were averaged to establish the overprint direction. The angular differences between the average overprint and all stable endpoint magnetizations were plotted on a histogram such as Figure 8. The histograms showed two families of directions with negligible overlap, so the families could be separated by a cutoff point centered in the saddle of the histogram. This procedure was applied to sites 1–4, where overprinting was appreciable, and resulted in the rejection of 24% of the thermally cleaned specimens (Table 2).

Site 5, collected in basaltic breccia, yielded widely dispersed endpoint magnetizations. Individual specimens typically consisted of several angular basalt fragments in a chlorite-rich, comminuted matrix. Thermal cleaning to 550°C usually isolated well-defined endpoint magnetizations, although cores collected within a meter of each other yielded dissimilar directions. We conclude that this site, which is a volcanogenic sedimentary deposit, did not acquire a coherent magnetization at the time of deposition. Instead, the basalt fragments acquired their magnetizations before coming to rest. Therefore results from site 5 are omitted from Table 2 and the calculation of paleolatitudes.

ROCK MAGNETISM

Two or three specimens were selected from each site for the measurement of Curie temperatures and the determination of isothermal remanence (IRM) saturation. Curie temperatures were measured in an automatic thermobalance, in which the specimens were heated in 0.3–0.4 T direct fields. Heating in argon or nitrogen at 1 atmosphere pressure gave best results because the balance was disturbed by gas streaming from secondary minerals when heatings were attempted in vacuum.

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Summary of Paleomagnetic Results From the Nikolai Greenstone, Central Alaska

Flow	$N(N_c)$	Treatment	R	k	α_{95}	I	D	I_c	D_c	θ	ϕ	A_{95}
						Site 1						
1	6(8)	527°–606°	5.9278	69	8.1	−37.2	7.3	−47.7	46.1	−8.9	352.8	
2	8(8)	586°–612°	7.7371	27	10.9	−42.2	41.2	−29.8	71.4	−6.1	326.2	
3	6(8)	608°–637°	5.7438	20	15.5	−38.0	47.2	−23.5	71.4	−2.8	324.6	
4	8(8)	606°–652°	7.9887	620	2.2	−33.6	35.9	−27.1	60.9	−0.5	334.7	
6	5(8)	608°–609°	4.9755	163	6.0	−28.3	35.1	−24.0	56.2	3.0	338.3	
7	8(8)	600°–637°	7.9706	238	3.6	−39.9	32.7	−33.2	64.5	−5.3	333.1	
8	7(8)	610°–629°	6.9501	120	5.5	−40.9	37.3	−31.2	68.1	−5.5	329.5	
9	8(8)	609°–635°	7.9020	71	6.6	−34.7	27.1	−33.4	56.3	−2.3	340.3	
10	8(8)	585°–618°	7.8959	67	6.8	−29.1	2.0	−46.1	32.8	−4.2	3.8	
Mean	9		8.7857	37	8.5	−36.9	29.3	−33.4	59.7	−3.7	338.1	8.5
						Site 2						
2	7(8)	546°–553°, 40mT	6.9437	107	5.9	−43.6	79.4	−44.2	146.4	−46.8	295.5	
3	3(8)	551°–574°, 40mT	2.9807	104	12.2	−35.8	73.3	−48.2	134.6	−45.4	275.3	
4	7(9)	40mT	6.8487	40	9.7	−37.4	74.1	−47.7	137.1	−46.1	272.1	
5	8(8)	40mT	7.9605	177	4.2	−31.9	95.3	−29.5	135.2	−33.5	267.3	
6	7(8)	550°–552°, 40mT	6.8741	48	8.8	−29.6	116.8	−11.8	140.9	−26.2	257.2	
8	8(8)	549°–552°, 40mT	7.8146	38	9.1	−30.6	100.6	−24.8	135.5	−31.0	265.7	
9	4(8)	550°–555°, 40mT	3.9467	56	12.4	−31.4	107.2	−19.9	138.8	−29.6	261.1	
Mean	7		6.7976	30	11.3	−35.4	93.1	−32.4	138.4	−37.1	265.1	7.7
						Site 3						
2	4(8)	596°–649°	3.9203	38	15.2	−45.0	60.8	−35.0	89.1	−16.8	313.4	
3	2(8)	619°–621°	1.9202	13	77.9	−36.4	66.5	−25.6	86.6	−10.5	312.9	
4	6(8)	551°–618°, 40mT	5.8046	26	13.5	−16.5	69.6	−7.9	76.9	2.3	317.2	
5	6(8)	600°–630°	5.8846	43	10.3	−10.9	72.8	−1.4	76.4	5.4	316.2	
6	3(8)	605°–609°	2.9231	26	24.7	−28.1	69.6	−17.4	83.6	−5.0	313.4	
7	5(9)	599°–609°	4.9683	126	6.8	−43.3	70.4	−29.3	94.0	−15.7	307.3	
Mean	6		5.8409	31	12.1	−30.1	68.6	−19.6	84.0	−6.7	313.4	8.2
						Site 4						
1	2(8)	523°–537°	1.9952	207	17.4	−4.9	89.8	7.3	90.4	3.1	302.1	
2	4(8)	519°–525°	3.9644	84	10.1	−12.2	102.9	−5.6	97.9	−6.0	298.3	
3	7(8)	499°–526°	6.9653	173	4.6	−27.2	88.8	−11.7	78.5	−0.2	317.0	
4	5(6)	513°–526°	4.9885	347	4.1	−13.3	102.7	−6.4	97.1	−6.1	299.3	
5	8(8)	487°–525°	7.9843	445	2.6	−17.0	104.5	−5.3	97.8	−5.9	298.4	
6	6(8)	449°–525°	5.7451	20	15.5	−4.9	91.0	−9.4	95.0	−6.5	301.8	
7	6(7)	399°–451°	5.9101	56	9.1	2.3	81.6	−8.0	83.1	−0.5	312.1	
9	5(8)	499°–520°	4.8180	22	16.7	−8.2	112.0	−9.6	107.3	−12.0	290.8	
10	7(8)	443°–525°	6.8360	37	10.1	−11.2	109.0	−10.7	103.1	−10.7	294.8	
Mean	9		8.8619†	58†	6.8†	−10.9	98.1	−6.7	94.5	−5.0	301.7	6.1
						Site 5‡						
						Site 6						
1	7(8)	549°–574°	6.8543	41	9.5	7.5	92.1	−18.4	98.0	−12.0	302.8	
2	7(8)	549°–599°	6.8575	42	9.4	−7.4	63.5	−50.3	92.1	−28.3	319.1	
3	8(8)	547°–599°	7.7853	33	9.9	−15.1	81.4	−39.0	114.2	−30.4	294.3	
4	6(7)	562°–575°	5.7518	20	15.3	−9.6	95.9	−23.9	115.9	−22.7	287.8	
5	7(8)	553°–576°	6.6050	15	16.0	5.0	96.4	−16.2	102.7	−13.1	298.0	
6	5(7)	553°–575°	4.9522	84	8.4	−4.2	85.0	−30.7	104.6	−21.3	300.3	
Mean	6		5.8268	29	12.7	−4.0	85.8	−29.9	104.9	−21.6	300.3	10.2
						Site 7						
1	8(8)	571°–604°	7.9584	168	4.3	−40.4	79.1	−50.3	121.2	−41.4	293.2	
2	8(8)	550°–622°	7.7430	27	10.8	−6.6	97.0	−15.3	104.9	−13.7	295.9	
3	8(8)	551°–597°	7.3077	10	18.3	−36.7	124.4	−20.8	145.8	−32.3	256.6	
4	6(6)	574°–655°	5.8112	26	13.3	−54.7	8.3	−79.9	259.9	−60.1	77.0	
5	7(8)	549°–605°	6.4042	10	20.0	−16.9	93.8	−25.2	109.6	−20.6	294.2	
6	8(8)	563°–600°	7.9072	75	6.4	−38.2	81.2	−50.7	111.5	−37.3	302.6	
7	8(8)	563°–585°	7.9810	368	2.9	−11.2	110.6	−13.4	117.8	−18.4	283.5	
8	8(8)	565°–599°	7.8645	52	7.8	−0.2	134.2	8.3	131.9	−13.7	265.5	
9	7(8)	549°–551°	6.7208	21	13.3	18.4	110.2	11.5	101.6	0.0	292.8	
Mean	9		7.5111†	5†	24.5†	−23.5	99.9	−27.4	119.0	−30.2	287.6	23.9

N_c is number of specimens collected; N is number of specimens averaged; Treatment refers to temperature range of thermal demagnetization in °C or applied alternating field in mTesla; R is resultant vector; k is Fisher precision parameter; α_{95}, 95% confidence limit for directions in degrees; I and D are inclination and declination without corrections for tilt of bedding, in degrees; I_c and D_c are inclination and declination of tilt-corrected directions, in degrees; θ and ϕ are latitude and east longitude of virtual geomagnetic pole, in degrees; and A_{95} is 95% confidence limit of mean pole.

*Sites from Figure 3. †Applies to corrected directions only. ‡Results omitted; see text.

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Rock magnetism of selected samples

Site (From Figure 3)	Flow	MDF, mT	T_B		Curie Temperature T_C	Pyrrhotite
			Overprint	Characteristic		
1	1	95	20°–530°	300°–680°	565°	
1	6	37	20°–580°	550° → 665°*	570°	
1	9	100	20°–610°	620°–680°	570°	
2	2	100	20°–400°	300° → 590°*	579°	X
2	5	100	20°–200°	200° → 590°*	575°	X
3	2	100	20°–490°	500° → 670°*	n.d.	
3	5	30	20°–600°	580°–670°	565°	X
4	1	85	20°–380°	190°–565°	570°	
4	3	55	20°–490°	490°–570°	n.d.	
6	2	100	20°–200°	200°–640°	565°	X
6	6	45	20°–200°	300°–680°	n.d.	
7	6	100	20°–300°	300°–680°	n.d.	
7	8	100	20°–500°	200°–680°	565°	trace
7	9	65	20°–400°	300°–680°	555°	

MDF is median destructive field obtained from AF demagnetization of NRM; T_B is the range of blocking temperatures derived from vector analysis of thermal demagnetization of NRM in °C; T_C are the Curie temperatures obtained from strong field thermomagnetic curves, in °C; X is the presence of pyrrhotite indicated by shape of thermomagnetic curve; n.d. means not determined.

*Highest temperature reached in experiment, although greater than 20% of NRM remained after last heating step.

The strong field method of determining Curie temperatures is effective for titanomagnetite compositions, but the method cannot detect hematite in the presence of even small quantites of titanomagnetite. The acquisition of IRM, however, provides an effective method for detecting hematite mixed with appreciable amounts of magnetite. IRM curves were generated by placing the specimen in progressively higher direct magnetic fields, up to the 0.7 T capacity of our electromagnet.

As revealed by the thermomagnetic determinations, most specimens gave Curie temperatures in the range 555°–579°C, indicating the presence of titanium-poor titanomagnetite (Table 3). Curie temperatures were indeterminable for four specimens because the strong field magnetization had low intensity and was dominated by the bulk paramagnetism of the specimens. A few specimens yielded heating curves having sharp increases in magnetization between 150°–165°C (Figure 9). This "dorsal fin" on the heating curve indicates the presence of minor pyrrhotite, for compostions ranging from Fe_9S_{10} toward FeS [*Schwarz*, 1975]. Because such pyrrhotite compositions are antiferromagnetic at room temperature, the mineral does not contribute to the NRM.

All specimens give IRM curves suggesting the presence of hematite, because all fail to reach saturation (Figures 10*a* and 10*b*). Curves showing appreciable acquisition of IRM in applied fields below 0.15 T and which flatten beyond 0.25 T indicate magnetite is present as well (Figure 10*b*). The detection of hematite is not surprising, since the thermal demagnetization of NRM revealed blocking temperatures as high as 680°C, the Neel point of hematite. Also, median destructive fields, as determined by AF demagnetization, were frequently greater than 100 mT, which is a characteristic of hematite (Table 3).

The presence of magnetite-hematite combinations in all test specimens was confirmed by microscopic examination of polished sections. The sections were examined at × 1050 magnification in reflected light, transmitted through oil. In general, the specimens contain euhedral or cruciform magnetite crystals, which exhibit two high-temperature (>600°C) oxidation features. Parent crystals exhibit a trellis pattern, associated with the "exsolution" of ilmenite from original titanomagnetite. In turn, the ilmenite-magnetite lamellae have partially altered to hematite plus rutile, by the process of pseudomorphic oxidation [*Haggerty*, 1976]. Another common feature of these basalts is abundant microcrystalline hematite that is associated with relict magnetite crystals, either as halos around grains or as pit fillings and veinlets within grains. This form of hematite, which imparts a reddish tinge to the basalt, probably formed during original cooling of the flows. In describing the Triassic flows of the Clearwater Mountains, *Smith* [1981] noted that *Surdam*'s [1968] description of hematitic flows of the Karmutsen Formation (Vancouver Island) applied to the thin, amygdaloidal flows of the Clearwaters as well. Because the hematite stain does not cross flow boundaries, and aquagene tuffs of the Karmutsen contain little or no hematite-rimmed iron oxides, *Surdam* [1968] concluded that

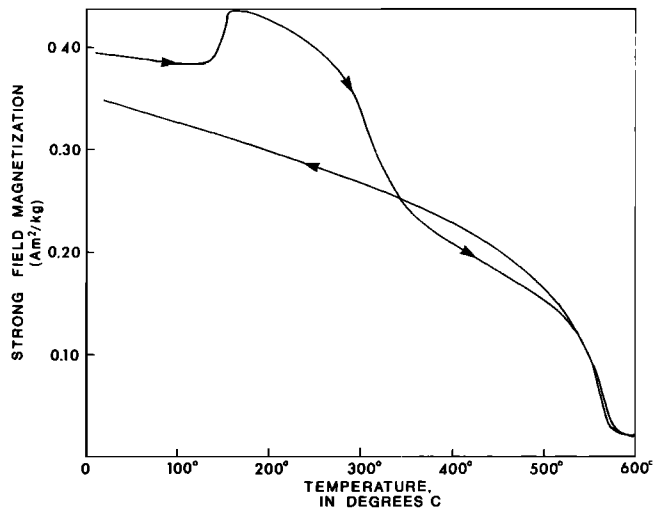


Fig. 9. Strong field (0.25 Tesla) magnetization induced as a function of temperature in a specimen from site 2, flow 2 (Figure 3; Tables 1–3). "Dorsal fin" on heating curve indicates the presence of pyrrhotite.

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

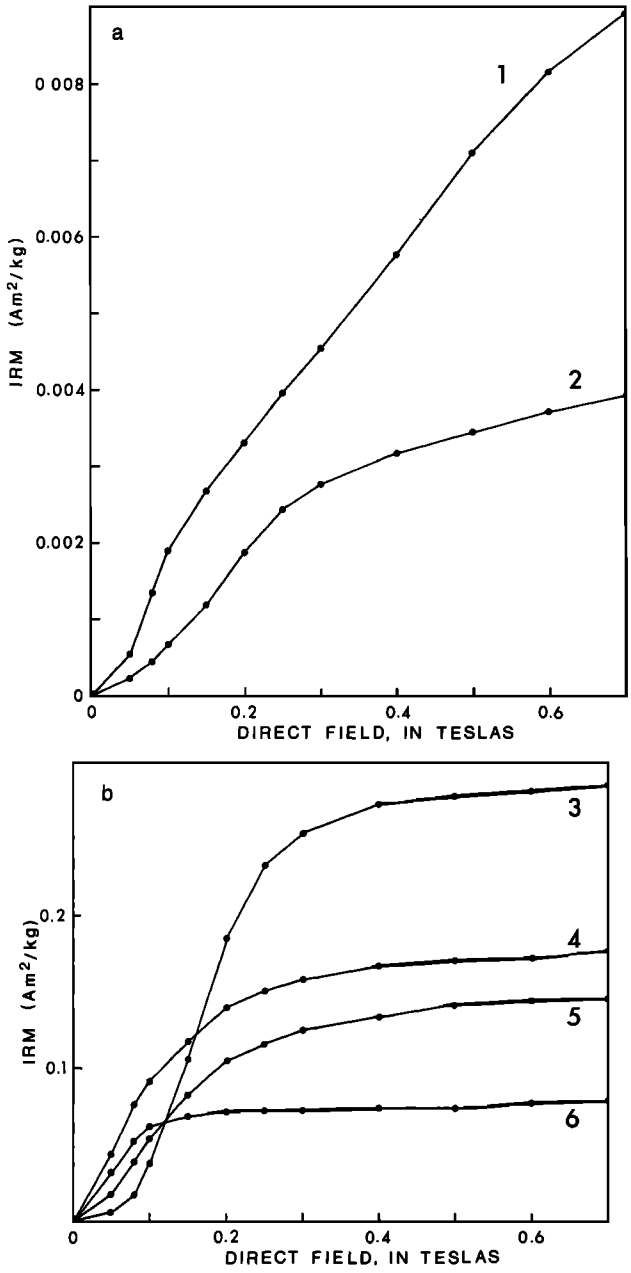


Fig. 10. Acquisition of isothermal remanent magnetization (IRM) in direct fields up to 0.7 T. (*a*) Specimens dominated by hematite: (1) site 3, flow 2, (2) site 7, flow 6. (*b*) Specimens containing magnetite and hematite: (3) site 2, flow 2, (4) site 1, flow 1, (5) site 3, flow 5, (6) site 1, flow 9. See Figure 1 and Tables 1–3.

the hematite was not the product of regional metamorphism. Instead, the hematite was interpreted as being formed by hot, oxygen-rich gases acting on iron oxide grains during crystallization and cooling of the flows.

Interpretation of Paleomagnetic Results

Mean directions that were calculated from thermally cleaned and selected data (Table 2) are plotted in Figure 11 to show the within-site angular dispersion. The selected directions, which are not corrected for tilt of the bedding, are contrasted with the mean overprinted direction that was established by vector subtraction of the thermal demagnetization results from sites 1–4. If the selection procedure had failed to remove the overprint entirely, as would occur if there was appreciable overlap of the blocking temperature distributions of the primary and overprinted magnetizations, then flow means would be streaked along a great circle passing through the overprinted direction. Results from sites 1, 2, and 4 are not streaked toward the overprint, indicating that the secondary magnetization has been removed successfully. In contrast, flow means from site 3 trend along a great circle through the secondary component, suggesting incomplete removal of the spurious remanence. We consider the contribution of the overprint to the total magnetization of these six flows to be minor and negligible when combined with the 40 flows from the other sites.

When corrected for tilt, the site mean directions cluster to the east with shallow and intermediate upward inclinations (Figure 12), but the tilt corrections do not change the angular dispersion significantly. The inconclusive fold test is in part due to the generally east-west strikes of the bedding and to localized rotations which introduce large variations in declination. Whether the variations are the result of noncylindrical folding or localized block rotations, the errors will not affect the inclinations. This is demonstrated by the distribution of tilt-corrected site means, in which the variation in declination is considerably greater than the spread in inclination.

Thermal demagnetization of the NRM indicates that the selected, stable endpoint magnetization is carried by hematite and to a lesser extent by magnetite because optimum cleaning temperatures usually exceeded 600°C. However, magnetite is probably the dominant magnetic constituent at site 4, where the blocking temperature distribution does not exceed 585°C, the Curie temperature of magnetite. Some demagnetization curves, for example, Figure 6, show sharp decreases in intensity near 585°C in addition to total unblocking temperatures near 680°C, suggesting that the selected endpoint remanence is shared by magnetite and hematite. In their study of the Karmutsen Formation, *Yole and Irving* [1980] isolated a high-temperature component (x magnetization) with thermal demagnetization properties similar to those of Figure 6. The x component is the basis for *Yole and Irving*'s [1980] paleolatitude determination of the Karmutsen.

We believe that the selected magnetization of the Nikolai was acquired during extrusion and cooling of the flows. Although the remanence is partly thermoremanent magnetization (TRM), the more important contribution is probably thermochemical remanent magnetization (TCRM) acquired when high-temperature oxidation produced hematite and magnetite at the expense of titanomagnetite. This high degree of oxidation, which is common during extrusion of subaerial lavas, is observed in the Nikolai and in the Karmutsen as well [*Surdam*, 1968]. As the selected magnetization is carried by magnetic domains having exceptionally high thermal stability, the magnetization has survived regional metamorphism which albitized original feldspars and altered the groundmass to chlorite. Although our results from the Healy and Mount Hayes quadrangles are not strengthed by a significant fold test or the consistency of normal and reversed directions (all flows have the same polarity), the virtual geomagnetic poles are very similar to results from the Nikolai Greenstone of McCarthy [*Hillhouse*, 1977] where such tests are available (Figure 13). In particular, the McCarthy study recorded an ancient polarity reversal with antipodal groups of field directions, indicating that a dipole magnetization free of overprints had been measured. We believe that our distribution of sites in the Tangle and Slana River subterranes

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

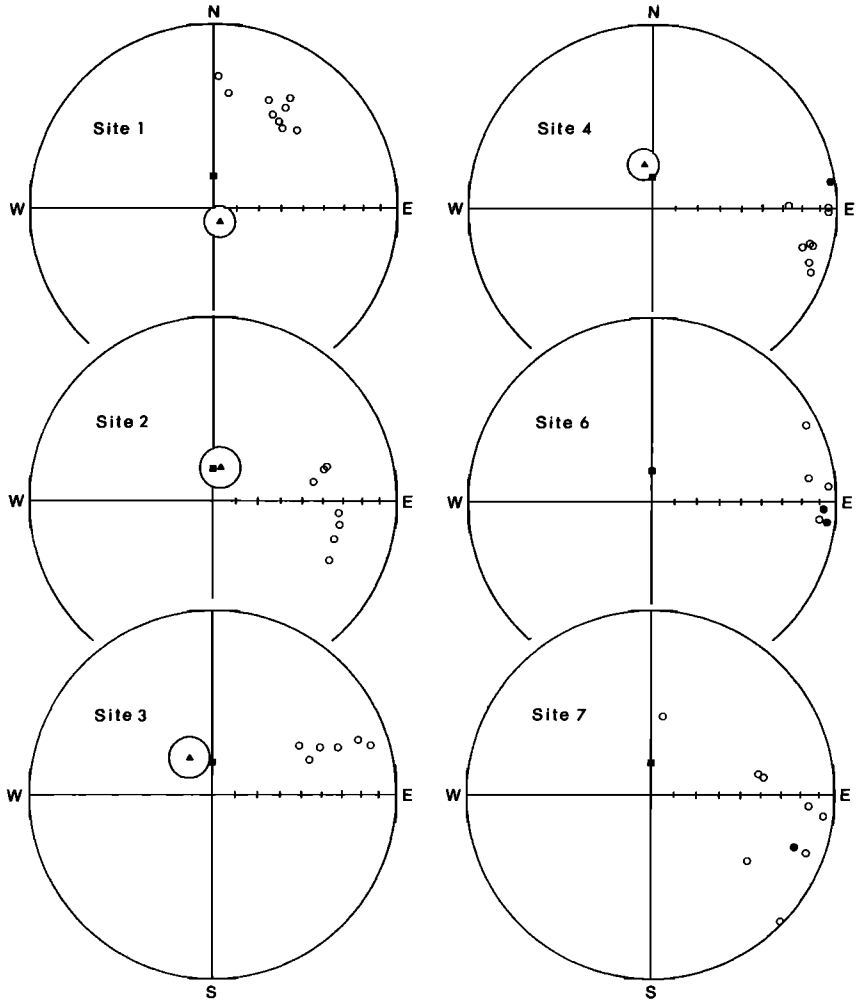


Fig. 11. Mean field directions (small circles) from flows before being corrected for tilt of the bedding. Mean overprint directions as determined by vector subtraction are shown by triangles within 95% confidence circles. Square denotes present axial dipole field. Solid symbols are in lower hemisphere; open symbols are in upper hemisphere of equal area projection. See Figure 3 and Table 1 for site locations.

represents a variety of levels within the extremely thick sequence of Nikolai flows and therefore adequately samples secular variation of the ancient geomagnetic field.

Using the assumption of a geocentric, axial dipole field, we calculated the mean Triassic paleolatitude of Wrangellia from the Healy–Mount Hayes results. In doing so, we used inclinations only, because localized rotations have introduced large variations in declination. Routine calculation of a paleomagnetic pole as a mean virtual geomagnetic pole (VGP) would result in inordinately poor confidence limits on the mean paleolatitude. Therefore inclinations of individual flow means were transformed to virtual geomagnetic latitudes (VGL) and then a mean paleolatitude and 95% confidence limit were calculated by the method of *McFadden and Reid* [1982]. The resultant mean paleolatitude is 13.9° (α_{95} = 3.8°).

Choosing the polarity of the results and hence the proper hemisphere of the paleolatitude has been debated since the first results from Wrangellia were obtained. In their early study of the Karmutsen Formation, *Irving and Yole* [1972] preferred an origin in the southern hemisphere because this choice required a smaller rotation of Vancouver Island. Placing the Karmutsen originally in the northern hemisphere would imply a lesser northward translation of Vancouver Island but would require a subsequent rotation of approximately 180° about a vertical axis to satisfy the observed paleodeclinations. Paleodeclinations from a fragment of Wrangellia in Hells Canyon (Oregon-Idaho) are consistent with this view [*Hillhouse et al.*, 1982]. Choosing the polarity on the basis of paleodeclinations is more ambiguous for the Alaskan fragment of Wrangellia, because the paleofield azimuths from the McCarthy, Healy, and Mount Hayes areas are highly variable (Figure 13). *Panuska and Stone* [1981] attempted to resolve the ambiguity at McCarthy by collecting two sites within Pennsylvanian and Lower Permian volcaniclastic rocks that underlie the Nikolai Greenstone. The resultant field directions, obtained from 21 specimens, are similar to the eastward-upward directions reported by *Hillhouse* [1977] from the Nikolai Greenstone. (Note that the field directions given in Table 1 of *Panuska and Stone* [1981] were inverted 180° from the original measurements (K. P. Kodama, written communication, May 1983).) Panuska and Stone concluded that Wrangellia originated in the northern hemisphere primarily during an epoch of reversed polarity, because the upper Paleozoic rocks were deposited during the Permian-Carboniferous reversed interval [*Irving and Pullaiah*, 1976]. We consider their assignment of polarity to the Nikolai results to be tentative because samples of the upper Paleozoic volcaniclastic rocks may have been remagnetized by the large, nearby

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

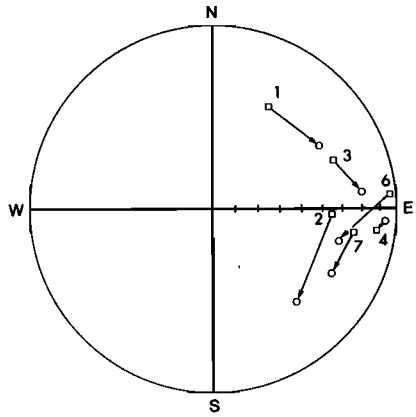


Fig. 12. Site mean directions before (squares) and after (circles) correction for tilt of the bedding. Site numbers are keyed to Table 1. All directions lie in upper hemisphere of equal area projection.

gabbro pluton that fed the Triassic basalt flows. Until the Pennsylvanian and Permian pole positions of Wrangellia are confirmed by a broader distribution of samples, we consider the matter open to debate and prefer to retain the southern option as a possibility.

Paleolatitude determinations are remarkably similar from Triassic rocks throughout Wrangellia from Alaska to Oregon. For comparison with the Healy–Mount Hayes paleolatitude (13.9 ±3.8°), we calculated means from paleomagnetic results obtained at McCarthy, Vancouver Island, and Hells Canyon. In each case, VGL's corresponding to a single flow or site were averaged and given 95% confidence limits according to the method of *McFadden and Reid* [1982]. As in the present study, paleodeclinations were ignored. Fifty lava flows from the Nikolai Greenstone at McCarthy yielded a paleolatitude of 10.4° ± 2.7° [*Hillhouse*, 1977]. When the results from two studies [*Yole and Irving*, 1980; *Schwarz et al.*, 1980] of the Karmutsen Formation of Vancouver Island are combined, the

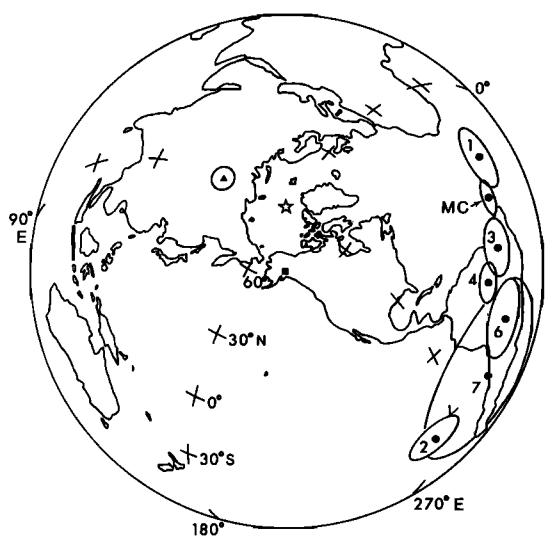


Fig. 13. Equal area projection centered on Mount Hayes quadrangle (square) showing mean virtual geomagnetic poles (dots) of sites 1–7, excluding 5, and 95% confidence limits (ellipses) from Triassic basalt in south-central Alaska. MC indicates Nikolai Greenstone at McCarthy [*Hillhouse*, 1977]. Triangle is Triassic reference pole with 95% confidence limit from nuclear North America (See Table 4). Present geographic pole is indicated by a star.

TABLE 4. Triassic and Early Jurassic Paleomagnetic Poles From Nuclear North America

	Locality		Unit VGP		Mean Paleomagnetic Pole			Reference
	Latitude, °N	Longitude, °E	Latitude, °N	Longitude, °E	Latitude, °N	Longitude, °E	α_{95}, deg	
Newark basin, New Jersey Watchung basalts and sediments of Newark Group	40.5	285.1			62.7	104.8	4.5	*Opdyke* [1961]*
Deerfield basin, Massachusetts Holyoke Lava Member and Granby Tuff of Meriden Formation (of Newark Group)	42.0	287.3			55.5	91.1	4.0	*Irving and Banks* [1961]*
Hartford basin, Connecticut	41.5	287.2			63.7	82.1	6.8	*de Boer* [1968]
Talcott lava			51	78				
Holyoke lava			60	83				
Hampden lava			73	92				
West Rock intrusive			63.2	86.0				*Smith and Noltimeier* [1979]
Mt. Carmel intrusive			66.7	87.3				
East Rock intrusive			67.3	70.4				
Fundy basin, Nova Scotia	45	295			72.2	101.3	21.6	*Carmichael and Palmer* [1968]
North Mountain basalt (lower part)			67	67				
North Mountain basalt (upper part)			71	149				
Shelburne dike			69	98				*Larochelle and Wanless* [1966]

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Moab, Utah Kayenta Formation	38.5	250.4			61.9	74.4	7.1, 3.7†	*Steiner and Helsley* [1974]
Lucianna Mesa, New Mexico Upper normal interval of Redonda Member and long reversed interval of unnamed shale member of Chinle Formation	35	256			62.5	89.0	5	*Reeve and Helsley* [1972]
Manicouagan, Quebec Intrusive rocks	51.4	291.3			59.7	88.5	4.6	
Site 2			50.5	97.4				
Site 7			61.4	111.3				
Site 6			62.5	91.3				
Site 8			47.3	56.4				*Robertson* [1967]
Site 3			54.1	91.4				
Site 4			61.6	97.7				
Site 1			62.5	72.3				
Site 2			60.8	89.1				
Site 3			65.3	125.0				
Site 4			56.8	111.8				
Site 5			57.2	75.5				
Site 6			59.9	90.1				*Larochelle and Currie* [1967]
Site 7			58.9	82.5				
Site 9			54.9	83.2				
Site 10			64.7	109.4				
Site 8			55.8	66.4				
Site 12			60.9	74.7				
Dolores River, Colorado Moenkopi Formation	38.6	251.1			54.2	102.9	4.0	
Interval N5			59.9	83.9				
Interval R5			53.2	95.9				
Interval N4			55.9	104.6				
Interval R4			53.5	105.2				
Interval N3			54.3	108.9				*Helsley and Steiner* [1974]
Interval R3B			55.1	113.6				
Interval R3A			48.6	113.7				
Interval R2			51.8	96.5				
Interval R1			52.9	100.2				*Helsley* [1969]
Gray Mountain, Arizona Moenkopi Formation Normal and reversed intervals combined	35.8	248.5			57.3	98.5	1.4	*Purucker et al.* [1980]

Overall mean paleomagnetic pole: 61.4°N, 92.5°E, $\alpha_{95} = 4.6°$, $N = 9$.
*Recalculated from original data.
†Semiaxes of the elliptical error about the pole at the 95% confidence level.

21562026, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

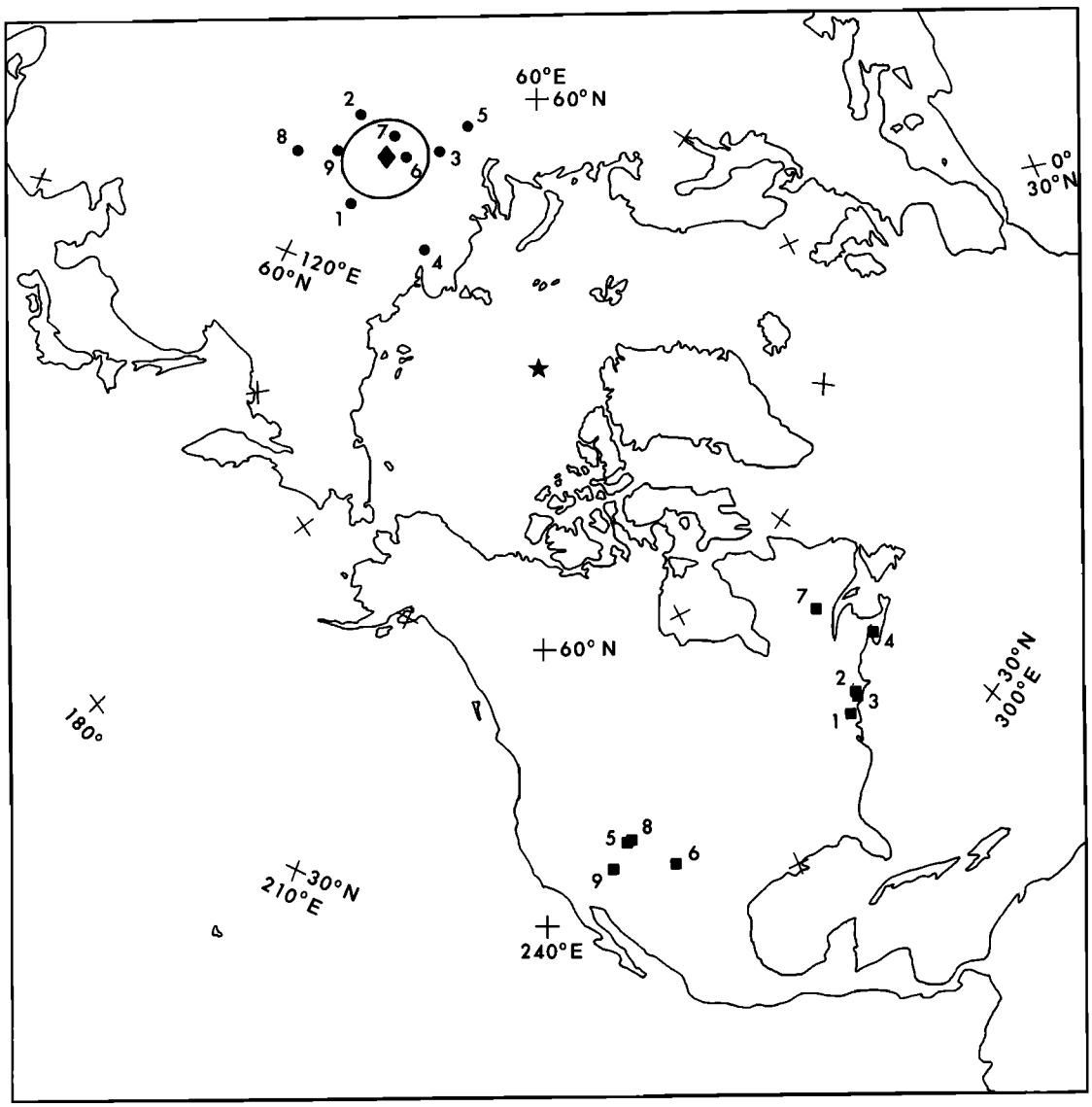


Fig. 14. Sample sites (squares) and paleomagnetic poles (dots) of Triassic and Early Jurassic age from nuclear North America. Numbers are keyed to Table 4. Mean pole is indicated by a diamond along with its 95% confidence circle. Star denotes north geographic pole.

resultant paleolatitude is 17.3° ±2.0°. The Seven Devils Group of Hells Canyon, Oregon and Idaho, yielded 17.5° ±4.1° [*Hillhouse et al.*, 1982].

NORTHWARD DISPLACEMENT OF WRANGELLIA

The determination of latitude differences between the original and present positions of Wrangellian fragments relative to the North American craton requires a reference paleomagnetic pole that is valid for Karnian and Ladinian time, the age of the Nikolai Greenstone. No North American rocks known to be this precise age have been sampled outside the tectonically disturbed belts of the Cordillera. The only recourse is to combine reference data from rocks that are slightly younger and older than the Nikolai Greenstone. We have compiled North American paleomagnetic poles of Triassic and Early Jurassic age (Table 4, Figure 14), selecting studies from tectonically stable areas. Our list does not include studies lacking adequate demagnetization treatments. We chose only Upper Triassic and Lower Jurassic units from the Newark Group by eliminating units of Middle Jurassic age, according to $^{40}Ar/^{39}Ar$ dating by *Sutter and Smith* [1979]. After the remaining unit VGP's were sifted for duplications, means were calculated for each of four basins sampled.

Two studies [*Robertson*, 1967; *Larochelle and Currie*, 1967] of the Early Triassic Manicouagan intrusive rocks were combined; however, the combined set may contain repeated measurements from the same cooling units. Our compilation also includes selected results from southwestern redbeds, such as the Chinle, Kayenta, and Moenkopi Formations. The Nikolai Greenstone is clearly older than the Early Jurassic intrusive rocks of the Newark Group and younger than the Moenkopi Formation. Although the minute details of Triassic apparent polar wander are unknown, we assume that the mean of all Triassic and Early Jurassic results is the best approximation of a late Middle Triassic reference pole (61.4°N, 92.5°E, $\alpha_{95} = 4.6°$, $N = 9$). The length of the time window used in this mean is approximately 30 m.y.

On the basis of this mean pole, a Triassic paleolatitude grid was drawn for North America (Figure 15). Latitudes determined for each site on the stable part of the continent are

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

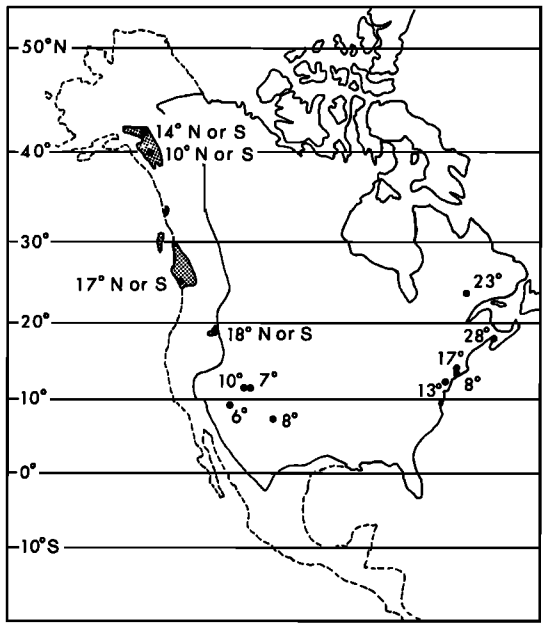


Fig. 15. Paleolatitude grid for North America in the Middle Triassic, as determined by the mean pole from nuclear North America (Table 4). Dashed coast line indicates region of Cordilleran terranes, including Wrangellia (stippled pattern). Paleolatitudes determined in Wrangellia and at the tectonically stable sites (see Figure 14) are shown for comparison with the mean paleolatitude grid. Latitudes of Wrangellia may lie in the northern or southern hemisphere (Table 5). From north to south the sites in Wrangellia are Healy–Mount Hayes, McCarthy, Vancouver Island, and Hells Canyon.

plotted to show how well the reference data compare with the mean grid. For comparison the fragments of Wrangellia are shown with their paleomagnetically determined latitudes. The Alaskan fragments of Wrangellia have significantly low latitudes, regardless of the hemisphere in which the rocks originated. In order to reach their present positions, the fragments have moved northward relative to the craton; a summary of the displacements is presented in Table 5. For example, the expected Triassic latitude of the Healy and Mount Hayes study area is 42.1°N (±4.6°). Given the observed paleolatitude of 13.9° (±3.8°) and assuming an origin in the northern hemisphere, the terrane's latitude has increased 28.2° ±6.0° in a coordinate system held fixed relative to North America. The displacement represented by the increase in latitude is 3100 km. For convenience we refer to the displacement as "northward drift," keeping in mind that the term does not refer to the present geographic coordinate system. The difference in latitude is a measure of the minimum displacement, because the original paleolongitude and subsequent path of the terrane are unknown. As shown in Table 5, the option of placing Wrangellia in the southern hemisphere suggests a minimum displacement of 6200 km for the Healy–Mount Hayes fragment.

HISTORY OF ACCRETION

From plate tectonic reconstructions [*Smith and Briden*, 1977], the possible source regions of Wrangellia are northwestern South America (Colombia, Ecuador, or northern Peru) or southern Tethys for the southern hemisphere option and North America (California) or southeast Asia for the northern option. Other possibilities are that Wrangellia was an island in the proto-Pacific or the margin of a continent, now fragmented and dispersed [*Nur and Ben-Avraham*, 1978]. The similar paleolatitudes of Wrangellian fragments suggest that the terrane was originally more compact than it is today and that the Alaskan fragments were displaced farther than were the fragments in British Columbia and Oregon. A less likely alternative is that Wrangellia has always been an elongate belt and was aligned along an east-west trend in the Triassic.

The zoogeographic affinities of Wrangellia were recently tested by a comparison of Norian bivalves from suspect terranes and the cratons that border the Pacific Ocean [*Newton*, 1983]. The study concluded that Wrangellia was located at low latitude in the eastern portion of the Triassic Pacific Ocean. Although the Wrangellian bivalves exhibit American affinities, knowledge of the faunas is too incomplete to resolve whether the terrane was located north or south of the equator. When a comparison is made between the Norian faunas of the Wrangell Mountains, Hells Canyon, the Peninsular terrane, and several cratonic sites in British Columbia and Nevada, the strongest correlation is between the Wrangell Mountains and Hells Canyon. This strong similarity supports the conclusion of *Jones et al.* [1977] that the two Wrangellian fragments,

TABLE 5. Paleolatitudes of Wrangellia Compared to the Triassic Position of North America

Area of Study	Source	N	Paleolatitudes $\pm\alpha_{95}$ Observed		Expected		Difference ±rms		Poleward Displacement, km
Healy and Mount Hayes quadrangles, Alaska	this study	46	13.9°N	±3.8°	42.1°N	±4.6°	28.2°	±6.0°	3100
			13.9°S				56.0°		6200
McCarthy quadrangle, Alaska	*Hillhouse* [1977]	50	10.4°N	±2.7°	40.0°N	±4.6°	29.6°	±5.3°	3300
			10.4°S				50.4°		5500
Vancouver Island, British Columbia	*Yole and Irving* [1980]; *Schwarz et al.* [1980]	37	17.3°N	±2.0°	25.1°N	±4.6°	7.8°	±5.0°	900
			17.3°S				42.4°		4700
Hells Canyon, Oregon and Idaho	*Hillhouse et al.* [1982]	11	17.5°N	±4.1°	19.0°N	±4.6°	1.5°	±6.2°	200
			17.5°S				36.5°		4000

N is the number of sites or flows sampled; α_{95} is the 95% confidence limit of mean paleolatitude in degrees; rms is the square root of sum of squares of confidence limits on observed and expected paleolatitudes in degrees.

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

now 1800 km apart, have a unique stratigraphy compared to the rocks of adjacent regions.

Definite geologic ties between Wrangellia and neighboring terranes are not evident until the last half of the Jurassic. Plutons of Late Jurassic and Early Cretaceous age, which knit together terranes of the Blue Mountains (Oregon) with the Seven Devils Group, give a paleolatitude indicating that poleward drift of southernmost Wrangellia was essentially complete 130 m.y. ago. Paleodeclinations of the plutons and remagnetized country rock suggest that the Blue Mountains province rotated clockwise subsequent to accretion [*Wilson and Cox*, 1980; *Hillhouse et al.*, 1982]. The timing of northward movement of Vancouver Island is constrained by paleomagnetic data [*Irving et al.*, 1980] from the Stikine terrane (Figure 1), which lies east of Wrangellia in central British Columbia [*Monger et al.*, 1982]. According to these results, the Stikine terrane was apparently 13° south of its present position relative to the craton, and separated from Wrangellia in the Late Jurassic. By Late Jurassic or Early Cretaceous time, Wrangellia (Vancouver Island) and the Stikine terrane were knitted together by the Coast Plutonic Complex. Inclinations determined from the middle Cretaceous Axelgold pluton in the Stikine terrane [*Irving et al.*, 1980] suggest that the amalgamated terranes remained about 13° south of their present position until Late Cretaceous or Early Tertiary time. The Axelgold pluton results suggest that Vancouver Island and the area east of it in the Stikine terrane have moved northward relative to the Seven Devils fragment of Wrangellia since the Cretaceous. This relative motion is consistent with the hypothesis that relative to Wrangellia's source region, poleward displacements of the more northern fragments are greater than the displacements of the southern fragments.

Paleomagnetic results from Triassic basalt of the Alexander terrane in southeastern Alaska [*Hillhouse and Gromme*, 1980] are difficult to reconcile with the accretionary scenario that has been outlined for British Columbia. The Alexander terrane lies between a fragment of Wrangellia to the west and the Stikine terrane to the east (Figure 1). Although Paleozoic latitudes of the Alexander terrane are 15° lower than expected [*Van der Voo et al.*, 1980], no northward drift of the terrane was detected in inclinations from the Upper Triassic Hound Island Volcanics, assuming a position in the northern hemisphere. Reconciliation of these observations with the results from British Columbia requires that the Stikine terrane be scissored northward between the Alexander terrane and the continent in the late Mesozoic or early Tertiary. Such a juxtaposition is possible because the region is broken by numerous, large-scale transcurrent faults. Alternatively, the true origin of the Alexander terrane may have been in the southern hemisphere, as suggested by the Peruvian affinity of a Norian bivalve from the terrane [*Newton*, 1983].

Northern Wrangellia and the Alexander terrane are overlapped by the Gravina-Nutzotin belt [*Berg et al.*, 1972], a sequence of highly deformed flysch and volcanic rocks of Late Jurassic and Early Cretaceous age (Figure 2). Flysch of equivalent age, also highly deformed but containing only minor volcanic rocks, occurs in structural contact between Wrangellia and the terranes of north central Alaska [*Csejtey et al.*, 1978, 1982; *Jones et al.*, 1982]. The Peninsular terrane, composed of Triassic bedded rocks and Early Jurassic plutons, lies south of the flysch belt, and at its northeast end the Peninsular terrane meets Wrangellia along a series of faults. Wrangellia and the Peninsular terrane are overlapped by Middle Jurassic and younger sedimentary rocks, which mark the amalgamation of the terranes into the "Talkeetna superterrane" of *Csejtey et al.* [1982].

The Triassic paleolatitude of the Peninsular terrane has not been determined, although the moderate similarity of Norian fauna of Wrangellia with fauna of the Peninsular terrane suggests that the terranes were not far apart in the Triassic [*Newton*, 1983]. Paleomagnetic data from Jurassic sedimentary rocks of the overlap assemblage imply that the amalgamated terranes were at a lower latitude relative to the craton in the Middle Jurassic [*Packer and Stone*, 1974; *Stone et al.*, 1982]. If the terranes were in the northern hemisphere by Jurassic time, then the data imply approximately 20° of northward drift relative to North America. However, *Panuska and Stone* [1981] favor the option of placing the amalgamated terranes in the southern hemisphere on the basis of minimum separation between the mean Jurassic paleomagnetic pole of the Peninsular terrane and the Triassic paleomagnetic pole of Wrangellia (assuming that Wrangellia originated in the northern hemisphere). This option implies a translational history for Wrangellia involving southward drift in the Early Jurassic followed by about 80° (9000 km) of post-Jurassic northward drift. As stated by Panuska and Stone, Wrangellia's polar wander curve, which is the basis for a Jurassic position in the southern hemisphere, is highly speculative because localized rotations within the terranes create large uncertainties in the azimuths to the mean paleomagnetic poles.

Anomalously low paleolatitudes have been determined from Upper Cretaceous and lower Tertiary sedimentary rocks of the Alaska Peninsula and the Matanuska Valley [*Stone et al.*, 1982]. The results imply that the southern terranes of Alaska underwent substantial northward drift (30°) relative to the craton during the Tertiary. If so, the latitude gap between northern and southern Alaska terranes must have closed in the region between the Matanuska Valley and the Denali fault in middle Tertiary time, because paleolatitudes of Paleocene volcanic rocks north of the Denali fault indicate that the northern region has remained fixed relative to the craton throughout the Tertiary [*Hillhouse and Gromme*, 1982; *Plumley and Coe*, 1982].

The geology of central Alaska provides very little evidence in support of *Stone et al.*'s [1982] contention that large-scale northward drift of Wrangellia and the Peninsular terrane occurred throughout the Tertiary. The proposed rate of drift would imply 3000 km of displacement since 50 m.y. ago. In the vicinity of the Denali fault, evidence of a closed ocean basin or subduction complex is absent in the Tertiary geologic record. Gentle, open folds are dominant in the Paleocene and lower Tertiary volcanic rocks which occur on both sides of the Denali fault. Although transcurrent motion of the southern terranes could close the implied latitude gap without requiring subduction and (or) compression of the region, Tertiary movements on the order of 3000 km are contradicted by mapping, which suggests much smaller offsets across the Denali fault or any other faults north of the Border Ranges fault [*Lanphere*, 1978; *Csejtey et al.*, 1982]. Furthermore, paleomagnetic results from lower Eocene volcanic rocks in the northern Talkeetna Mountains [*Hillhouse and Gromme*, 1983; *Hillhouse et al.*, 1984] indicate that the northwestern part of Wrangellia has moved little relative to the craton since 50 m.y. ago.

We prefer a middle Cretaceous collision between Wrangellia, as part of the previously assembled Talkeetna superterrane, and the Alaska mainland, as suggested by reconnaissance geologic mapping of the Talkeetna Mountains and the Central Alaska Range [*Csejtey et al.*, 1978, 1982; *Jones et al.*,

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1982]. The backstop for the collision would be the Nixon Fork and Yukon-Tanana terranes of central Alaska (Figure 1). The widespread and highly deformed upper Mesozoic flysch constitutes the obvious choice for the collision zone in which tremendous north-south shortening occurred. The original size and latitude of the flysch basin are poorly known, although in south-central Alaska the flysch was deposited on the leading edge of Wrangellia. Rocks of the Gravina-Nutzotin belt, which may be part of the flysch, extend as far south as Gravina Island (southeastern Alaska); part of the basin may have originated in this region. Fossil ammonites from the flysch are typical of northern North American fauna [*Jones*, 1973]; this fact contradicts tectonic models proposing an original position of the flysch basin far south of the Pacific Northwest.

The age of metamorphism, as inferred from isotopic studies, may relate to the timing of accretion in south-central Alaska. The Maclaren metamorphic belt [*Smith*, 1981], which primarily consists of argillite, high-grade schist, gneiss, and quartz diorite, borders the northern part of Wrangellia in the Clearwater Mountains. *Smith* [1981] bracketed the age of intense metamorphism of the Maclaren between 143 m.y. and 57 m.y. on the basis of K-Ar determinations from metamorphosed and unmetamorphosed plutons in the belt. The low-grade metamorphism (prehnite-pumpellyite facies) of the Wrangellian Triassic rocks may be related to the primarily late Mesozoic pulse of metamorphism that was observed in the Maclaren belt. *Silberman et al.* [1981] have obtained several potassium-argon ages from Nikolai Greenstone samples from the McCarthy area, which range from 91 to 131 m.y., all much younger than the age of rocks. The data form an isochron of 112 $\pm$11 m.y. when plotted on an initial argon diagram, and Silberman et al. interpret this reheating age as representing the final accretion of Wrangellia to the continent. They favored this hypothesis over plutonism to explain the resetting of ages because plutons of the appropriate age are rare in Wrangellia. *Turner and Smith* [1974] reported the following K-Ar ages from intrusive rocks in Wrangellia in the Susitna River region. A small quartz-diorite body in the Clearwater Mountains gave a hornblende age of 125.8 m.y. Granitic plutons in the Paleozoic volcanic rocks to the south gave Jurassic ages of 145 m.y., 143 m.y., and 173 m.y.

The data do not yet provide a simple answer to the question of when the metamorphism of western Wrangellia occurred or whether the metamorphism was related to the collision of the Talkeetna superterrane with the mainland. The magnetic overprint (Figure 7) is perhaps a manifestation of regional heating, but the age of the overprint is only loosely constrained. The magnetic directions have normal polarity and transform to poles that are generally consistent with the Late Cretaceous and younger apparent polar wander path of North America.

While Wrangellia and the Peninsular terrane moved northward in the Cretaceous, a number of detached crustal fragments, such as the Chulitna terrane [*Jones et al.*, 1980; *Csejtey et al.*, 1982] were thrust over and folded into the deformed flysch (Figure 2). Prior to 50 m.y. ago, the deformation within the collision zone was essentially complete, as isoclinal folds within the flysch basin were cut by plutons and overlapped by volcanic rocks. The Talkeetna thrust, which borders northern Wrangellia, was also intruded and overlapped by igneous rocks at this time [*Csejtey et al*, 1978]. These lower Tertiary volcanic rocks are considerably less deformed than are the upper Mesozoic flysch deposits. The myriad of faults in southern Alaska have modified the late Mesozoic collision zone in Tertiary time, but displacements of a magnitude (>500 km) detectable by paleomagnetic methods probably did not occur [*Csejtey et al.*, 1982].

The terranes of the Chugach Mountains, Prince William Sound, and Kodiak Island, however, show evidence of large-scale transport and accretion in the Tertiary. Lying south of the Border Ranges fault (Figure 2), the Chugach terrane is a highly deformed wedge of Upper Cretaceous turbidites with isolated complexes of pillow basalt and sheeted dikes [*Tysdal et al.*, 1977]. Paleolatitudes measured in the sedimentary rocks [*Stone et al.*, 1982] and in the volcanic and intrusive rocks [*Gromme and Hillhouse*, 1981] of the Chugach terrane are anomalously low, suggesting Tertiary displacement of 20°–30° northward relative to the craton. Similar displacements were inferred from a paleomagnetic study of highly deformed Paleocene volcanic rocks on the southern coast of Kodiak Island (Figure 1) in the Prince William terrane [*Plumley et al.*, 1982; 1983].

CONCLUSIONS

Paleomagnetic sampling of the Triassic Nikolai Greenstone has been extended 300 km northwest from McCarthy to the Healy quadrangle in southern Alaska. After thermal demagnetization, the basalt specimens reveal two components of magnetization: (1) a steep downward overprint that fails the fold test and (2) a characteristic magnetization that presumably was acquired during original cooling of the basalts. Inclinations measured from the 46 lava flows yield a mean paleolatitude of 13.9° (α_{95} = 3.8°). We prefer to leave open the question of the polarity of the data and hence the hemisphere of origin; however, preliminary paleomagnetic data of *Panuska and Stone* [1981] favor the northern hemisphere as the birthplace of Wrangellia.

Extensive sampling of the Triassic rocks of Wrangellia in Alaska, British Columbia, Oregon, and Idaho has consistently yielded paleolatitudes of 10°–17°. These mean paleolatitudes, which are derived from the literature and the new results presented here, are calculated from inclinations only, because localized rotations within Wrangellia have introduced large variations in declination. When compared to results from similar-aged rocks of stable North America, Wrangellian paleolatitudes are anomalously low, implying northward translation exceeding 3000 km for the northern part of the terrane. We favor a tectonic model whereby most of Wrangellia's northward drift occurred between the Late Triassic and Late Cretaceous. The terrane was broken apart and fragments were left scattered along the continental margin after accretion. The farthest-travelled Alaskan fragment, represented by paleomagnetic data from the McCarthy, Mount Hayes, and Healy quadrangles, probably reached mainland Alaska in the middle of the Cretaceous, when the intervening flysch basin was crumpled by the collision. The positions of Wrangellia's Alaskan fragments have been modified by Tertiary transcurrent faulting, but we consider the possible fault displacements to be minor, as indicated by paleomagnetic results from overlapping, early Tertiary volcanic rocks.

Acknowledgments. We thank David L. Jones, Norman J. Silberling, Donald H. Richter, and Warren J. Nokleberg for extensive discussions and for generous support during the field work.

REFERENCES

Berg, H. C., D. L. Jones, and D. H. Richter, Gravina-Nutzotin belt-Tectonic significance of an upper Mesozoic sedimentary and vol-

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

canic sequence in southern and southeastern Alaska, *U.S. Geol. Surv. Prof. Paper, 800-D*, D1–D24, 1972.
Carmichael, C. M., and A. C. Palmer, Paleomagnetism of the Late Triassic North Mountain Basalt of Nova Scotia, *J. Geophys. Res., 73*, 2811–2822, 1968.
Coney, P. J., D. L. Jones, and J. W. H. Monger, Cordilleran suspect terranes, *Nature, 288*, 329–333, 1980.
Csejtey, B., Jr., Tectonic implications of a Late Paleozoic volcanic arc in the Talkeetna Mountains, south-central Alaska, *Geology, 4*(1), 49–52, 1976.
Csejtey, B., Jr., W. H. Nelson, D. L. Jones, N. J. Silberling, R. M. Dean, M. J. Morris, M. A. Lanphere, J. G. Smith, and M. L. Silberman, Reconnaissance geologic map and geochronology, Talkeetna Mountains quadrangle, northern part of Anchorage quadrangle, and southwest corner of Healy quadrangle, scale 1:250,000, Alaska, *U.S. Geol. Surv. Open File Rep. 78-558A*, 62 pp., 1978.
Csejtey, B., Jr., D. P. Cox, R. C. Evarts, G. D. Stricker, and H. L. Foster, The Cenozoic Denali fault system and the Cretaceous accretionary development of southern Alaska, *J. Geophys. Res., 87*, 3741–3754, 1982.
de Boer, J., Paleomagnetic differentiation and correlation of the Late Triassic volcanic rocks in the central Appalachians (with special reference to the Connecticut Valley), *Geol. Soc. Am. Bull., 79*, 609–626, 1968.
Gromme, C. S., and J. W. Hillhouse, Paleomagnetic evidence for northward movement of the Chugach terrane, southern and southeastern Alaska, *U.S. Geol. Surv. Circ. 823-B*, B70–B72, 1981.
Haggerty, S. E., Opaque mineral oxides in terrestrial igneous rocks, *Short Course Notes Mineral. Soc. Am., 3*, Hg 101–Hg 300, 1976.
Helsley, C. E., Magnetic reversal stratigraphy of the Lower Triassic Moenkopi Formation of western Colorado, *Geol. Soc. Am. Bull., 80*, 2431–2450, 1969.
Helsley, C. E., and M. B. Steiner, Paleomagnetism of the Lower Triassic Moenkopi Formation, *Geol. Soc. Am. Bull., 85*, 457–464, 1974.
Hillhouse, J. W., Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy quadrangle, Alaska, *Can. J. Earth Sci., 14*, 2578–2592, 1977.
Hillhouse, J. W., and C. S. Gromme, Paleomagnetism of the Triassic Hound Island Volcanics, Alexander terrane, southeastern Alaska, *J. Geophys. Res., 85*, 2594–2602, 1980.
Hillhouse, J. W., and C. S. Gromme, Limits to northward drift of the Paleocene Cantwell Formation, central Alaska, *Geology, 10*, 552–556, 1982.
Hillhouse, J. W. and C. S. Gromme, Wrangellia in southern Alaska 50 m.y. ago, *Eos Trans. AGU, 64*, 687, 1983.
Hillhouse, J. W., C. S. Gromme, and T. L. Vallier, Paleomagnetism and Mesozoic tectonics of the Seven Devils volcanic arc in northeastern Oregon, *J. Geophys. Res., 87*, 3777–3794, 1982.
Hillhouse, J. W., C. S. Gromme, and B. Csejtey, Jr., Paleomagnetism of Early Tertiary volcanic rocks in the northern Talkeetna Mountains, *Stanford Univ. Publ. Geol. Sci., 18*, 111–114, 1984.
Irving, E., and M. R. Banks, Paleomagnetic results from the Upper Triassic lavas of Massachusetts, *J. Geophys. Res., 66*, 1935–1939, 1961.
Irving, E., and G. Pullaiah, Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of paleosecular variation in the Phanerozoic, *Earth Sci. Rev., 12*, 35–64, 1976.
Irving, E., and R. W. Yole, Paleomagnetism and kinematic history of mafic and ultramafic rock in fold mountain belts, *Publ. Earth. Phys. Branch Can., 42*, 87–95, 1972.
Irving, E., J. W. H. Monger, and R. W. Yole, New paleomagnetic evidence for displaced terranes in British Columbia, *Spec. Pap. Geol. Assoc. Can., 20*, 441–456, 1980.
Jones, D. L., Structural elements and biostraitigraphical framework of Lower Cretaceous rocks in southern Alaska, *Geol. J. Spec. Issue, 5*, 1–8, 1973.
Jones, D. L., N. J. Silberling, and J. W. Hillhouse, Wrangellia-A displaced terrane in northwestern North America, *Can. J. Earth Sci., 14*, 2565–2577, 1977.
Jones, D. L., N. J. Silberling, B. Csejtey, Jr., W. H. Nelson, and C. D. Blome, Age and structural significance of ophiolite and adjoining rocks in the Upper Chulitna District, south-central Alaska, *U.S. Geol. Surv. Prof. Pap. 1121-A*, A1–A21, 1980.
Jones, D. L., N. J. Silberling, H. C. Berg, and George Plafker, Map showing tectonostratigraphic terranes of Alaska, columnar sections, and summary description of terranes, scale 1:2,500,000, *U.S. Geol. Surv. Open File Rep. 81-792*, 20 pp., 1981.
Jones, D. L., N. J. Silberling, W. G. Gilbert, and P. J. Coney, Character, distribution and tectonic significance of accretionary terranes in the central Alaska Range, *J. Geophys. Res., 87*, 3709–3717, 1982.
Lanphere, M. A., Displacement history of the Denali fault system, Alaska and Canada, *Can. J. Earth Sci., 15*, 817–822, 1978.
Larochelle, A., and K. L. Currie, Paleomagnetic study of igneous rocks from the Manicouagan structure, Quebec, *J. Geophys. Res., 72*, 4163–4169, 1967.
Larochelle, A., and R. K. Wanless, The paleomagnetism of a Triassic diabase dike in Nova Scotia, *J. Geophys. Res., 71*, 4949–4953, 1966.
MacKevett, E. M., Jr., Geologic map of the McCarthy quadrangle, Alaska, scale 1:250,000, *U.S. Geol. Surv. Misc. Invest. Ser.*, Map I-1032, 1978.
McFadden, P. L. and A. B. Reid, Analysis of paleomagnetic inclination data, *Geophys. J. R. Astron. Soc., 69*, 307–320, 1982.
Monger, J. W. H., J. G. Souther, and H. Gabrielse, Evolution of the Canadian Cordillera: A plate-tectonic model, *Am. J. Sci., 272*, 577–602, 1972.
Monger, J. W. H., R. A. Price, and D. J. Tempelman-Kluit, Tectonic accretion and the origin of the two major metamorphic and plutonic welts in the Canadian Cordillera, *Geology, 10*, 70–75, 1982.
Newton, C. R., Paleozoogeographic affinities of Norian bivalves from the Wrangellian, Peninsular, and Alexander terranes, western North America, in *Pre-Jurassic Rocks in Western North American Suspect Terranes*, edited by C. H. Stevens, pp. 37–48, Society of Economic Paleontologists and Mineralogists, Los Angeles, 1983.
Nokleberg, W. J., N. R. D. Albert, P. L. Herzon, R. T. Miyaoka, and R. E. Zehner, Recognition of two subterranes within the Wrangellia terrane, southern Mount Hayes quadrangle, Alaska, *U.S. Geol. Surv. Circ. 823-B*, B64–B66, 1981.
Nokleberg, W. J., et al., Geological map of the southern part of the Mount Hayes quadrangle, Alaska, scale 1:250,000, *U.S. Geol. Surv. Open File Rep. 82-52*, 1982.
Nur, A., and Z. Ben-Avraham, Speculations on mountain building and the lost Pacifica continent, *J. Phys. Earth Suppl., 26*, S21–S37, 1978.
Opdyke, N. D., The paleomagnetism of the New Jersey Triassic: A field study of the inclination error in red sediments, *J. Geophys. Res., 66*, 1941–1949, 1961.
Packer, D. R., and D. B. Stone, Paleomagnetism of Jurassic rocks from southern Alaska, and their tectonic implications, *Can. J. Earth Sci., 11*, 976–997, 1974.
Panuska, B. C., and D. B. Stone, Late Paleozoic paleomagnetic data from Wrangellia: Resolution of the polarity ambiguity, *Nature, 293*, 561–563, 1981.
Plumley, P. W., and R. S. Coe, Paleomagnetic evidence for reorganization of the north Cordilleran borderland in late Paleozoic time, *Geol. Soc. Am. Abstr. Programs, 14*, 224, 1982.
Plumley, P. W., R. S. Coe, T. Byrne, M. R. Reid, and J. C. Moore, Paleomagnetism of the Kodiak Islands indicates northward latitudinal displacement, *Nature, 300*, 50–52, 1982.
Plumley, P. W., R. S. Coe, and T. Byrne, Paleomagnetism of the Paleocene Ghost Rocks Formation, Prince William terrane, Alaska, *Tectonics, 2*(3), 295–314, 1983.
Purucker, M. E., D. P. Elston, and E. M. Shoemaker, Early acquisition of characteristic magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona, *J. Geophys. Res., 85*, 997–1012, 1980.
Reeve, S. C., and C. E. Helsley, Magnetic reversal sequence in the upper portion of the Chinle Formation, Montoya, New Mexico, *Geol. Soc. Am. Bull., 83*, 3795–3812, 1972.
Richter, D. H., and J. T. Dutro, Jr., Revision of the type Mankomen Formation (Pennsylvanian and Permian), Eagle Creek area, eastern Alaska Range, Alaska, *U.S. Geol. Surv. Bull. 1395-B*, B1–B25, 1975.
Richter, D. H., W. N. Sharp, J. T. Dutro, Jr., and W. B. Hamilton, Geologic map of parts of the Mount Hayes A-1 and A-2 quadrangles, Alaska, scale 1:63,360, U.S. Geol. Surv. Misc. Invest. Ser., Map I-1031, 1977.
Robertson, W. A., Manicouagan, Quebec, paleomagnetic results, *Can. J. Earth Sci., 4*, 641–649, 1967.
Rohn, O., A reconnaissance of the Chitina River and the Skolai Mountains, Alaska, *U.S. Geol. Surv. Ann. Rep. 21st Part 2*, 399–440, 1900.
Schwarz, E. J., Magnetic properties of pyrrhotite and their use in applied geology and geophysics, *Geol. Surv. Can. Spec. Pap. 74-59*, 24 pp., 1975.
Schwarz, E. J., J. E. Muller, and K. R. Clark, Paleomagnetism of the

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Karmutsen basalts from southeast Vancouver Island, *Can. J. Earth Sci., 17*, 389–399, 1980.
Silberling, N. J., D. H. Richter, and D. L. Jones, Recognition of the Wrangellia terrane in the Clearwater Mountains and vicinity, south-central Alaska, *U.S. Geol. Surv. Circ. 823-B*, B51–B55, 1981*a*.
Silberling, N. J., D. H. Richter, D. L. Jones, and P. J. Coney, Geologic Map of the bedrock part of the Healy A-1 quadrangle south of the Talkeetna-Broxson Gulch fault system, Clearwater Mountains, Alaska, scale 1:63,360, *U.S. Geol. Surv. Open File Rep. 81-1288*, 1981*b*.
Silberman, M. L., E. M. MacKevett Jr., C. L. Connor, P. L. Klock, and G. Kalechitz, K-Ar ages of the Nikolai Greenstone from the McCarthy quadrangle, Alaska: The "docking" of Wrangellia, *U.S. Geol. Surv. Circ. 823-B*, B61–B63, 1981.
Smith, A. G., and J. C. Briden, *Mesozoic and Cenozoic Paleocontinental Maps*, 63 pp., Cambridge University Press, London, 1977.
Smith, J. G., and E. M. MacKevett, Jr., The Skolai Group in the McCarthy B-4, C-4, and C-5 quadrangles, Wrangell Mountains, Alaska, *U.S. Geol. Surv. Bull. 1274-Q*, 27 pp., 1970.
Smith, T. E., Regional geology of the Susitna-Maclaren River area, *Alaska Div. Geol. Geophys. Surv. Annu. Rep. 1973*, 3–6, 1974.
Smith, T. E., Geology of the Clearwater Mountains, south-central Alaska, *Alaska Div. Geol. Geophys. Surv. Geol. Rep. 60*, 72 pp., 1981.
Smith, T. E., and H. C. Noltimier, Paleomagnetism of the Newark trend igneous rocks of the north central Appalachians and the opening of the central Atlantic Ocean, *Am. J. Sci., 279*, 778–807, 1979.
Smith, T. E., and D. L. Turner, Geochronology of the Maclaren metamorphic belt, south-central Alaska: A progress report, *Isochron/West, 7*, 21–25, 1973.
Steiner, M. B., and C. E. Helsley, Magnetic polarity sequence of the Upper Triassic Kayenta Formation, *Geology, 2*, 191–194, 1974.
Stone, D. B., Triassic paleomagnetic data and paleolatitudes for Wrangellia, Alaska, *Alaska Div. Geol. Geophys. Surv. Geol. Rep. 73*, 55–62, 1982.
Stone, D. B., B. C. Panuska, and D. R. Packer, Paleolatitudes versus time for southern Alaska, *J. Geophys. Res., 87*, 3697–3707, 1982.
Stout, J. H., Geology of the Eureka Creek area, east central Alaska Range, *Alaska Div. Geol. Geophys. Surv. Geol. Rep. 46*, 32 pp., 1976.
Surdam, R. C., Origin of native copper and hematite in the Karmutsen Group, Vancouver Island, B. C., *Econ. Geol., 63*, 961–966, 1968.
Sutter, J. F., and T. E. Smith, $^{40}Ar/^{39}Ar$ ages of diabase intrusions from Newark Trend basins in Connecticut and Maryland: Initiation of central Atlantic rifting, *Am. J. Sci., 279*, 808–831, 1979.
Symons, D. T. A., Paleomagnetic notes on the Karmutsen Basalts, Vancouver Island, British Columbia, *Can. Geol. Surv. Pap. 71-24*, 11–24, 1971.
Turner, D. L., and T. E. Smith, Geochronology and generalized geology of the central Alaska Range, Clearwater Mountains, and northern Talkeetna Mountains, *Alaska Div. Geol. Geophys. Surv. Open File Rep. 72*, 11 pp., 1974.
Tysdal, R. G., J. E. Case, G. R. Winkler, and S. H. B. Clark, Sheeted dikes, gabbro, and pillow basalt in flysch of coastal southern Alaska, *Geology, 5*, 377–383, 1977.
Van der Voo, R., M. Jones, C. S. Gromme, G. D. Eberlein, and M. Churkin, Jr., Paleozoic paleomagnetism and the northward drift of the Alexander terrane, southeastern Alaska, *J. Geophys. Res., 85*, 5281-5296, 1980.
Wilson, D., and A. Cox, Paleomagnetic evidence for tectonic rotation of Jurassic plutons in Blue Mountains, eastern Oregon, *J. Geophys. Res., 85*, 3681–3689, 1980.
Yole, R. W., and E. Irving, Displacement of Vancouver Island: Paleomagnetic evidence from the Karmutsen Formation, *Can. J. Earth Sci., 17*, 1210–1288, 1980.

C. S. Gromme and J. W. Hillhouse, U.S. Geological Survey, Isotope Geology, Mail Stop 937, 345 Middlefield Road, Menlo Park, CA 94025.

(Received June 28, 1983;
revised January 12, 1984;
accepted January 17, 1984.)

21562202b, 1984, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB089iB06p04461 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License